



WSQ COURSE SLIDES

Basic Urban Farming with Hydroponics

Course Code: TGS-2025053916

Version 2.0 · 29 August 2026

Tertiary Infotech Academy Pte Ltd
UEN 201200696W

Root-zone control • safe operations • Singapore cases



Your Trainer



**GENERAL TRAINER PROFILE ·
COMPLETE BEFORE DELIVERY**

Name and preferred form of address

Relevant urban-farming / hydroponics experience

Training and Adult Education credentials

Site emergency and escalation role

Dr Alfred Ang



**COURSE LEAD · TERTIARY
INFOTECH ACADEMY**

WSQ trainer and courseware lead

Focus: mechanisms, measurements and evidence

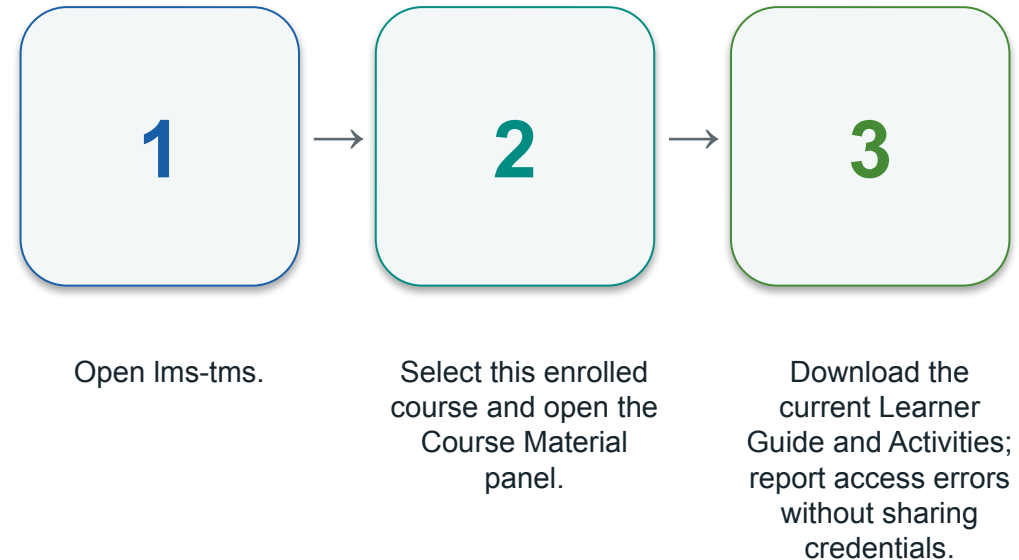
Ask about system selection and troubleshooting

Escalation: assessment and course administration

Download Course Material

ANCHOR: sign in → open enrolled course → download current files

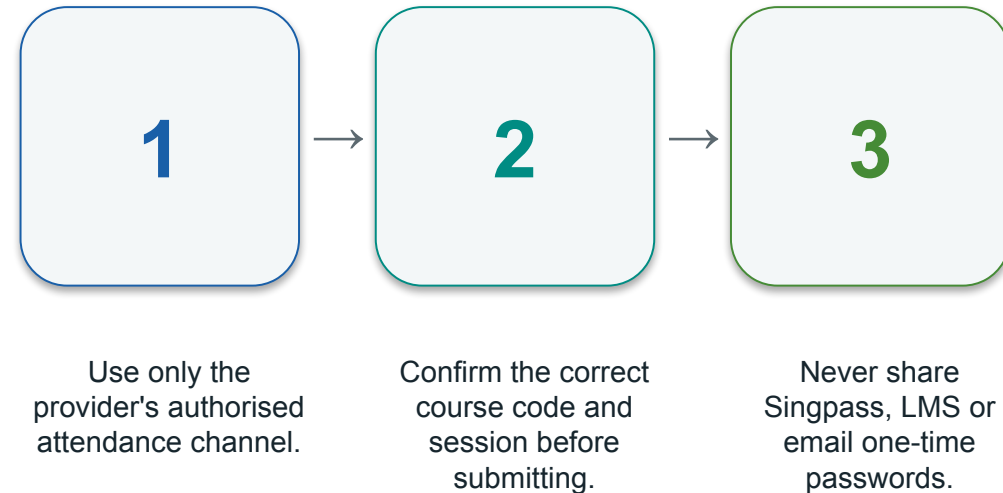
- Open lms-tms.tertiaryinfotech.com using your authorised learner account.
- Select this enrolled course and open the Course Material panel.
- Download the current Learner Guide and Activities; report access errors without sharing credentials.



Digital Attendance at Start

ANCHOR: identity → attendance event → confirmation

- Use only the provider's authorised attendance channel.
- Confirm the correct course code and session before submitting.
- Never share Singpass, LMS or email one-time passwords.



Practice Exam / Knowledge Check

ANCHOR: course WA-SAQ + observed PP are the assessed instruments

- This is a non-certification WSQ course; revision centres on K1–K4 and A1–A4.
- Use the trainer-provided hydroponics knowledge check and both activity evidence sheets.
- Optional Tertiary practice portal:
<https://exams.tertiaryinfotech.com/>

1 This · non-certification · course · revision · centres · K1–K4

2 trainer-provided · hydroponics · knowledge · check · both · activity

3 Optional · Tertiary · practice · portal · <https://exams.tertiaryinfotech.com/>

Course outcomes

ANCHOR: Perform hydroponic operations to SOP; apply WSH, food-safety, cleaning and maintenance controls.

- K1–K4 equipment, SOP, GAP/WSH, maintenance
- A1–A4 prepare, operate, comply, clean/maintain

1

K1–K4 · equipment · GAP/WSH · maintenance

2

**A1–A4 · prepare · operate · comply ·
clean/maintain**

Today's operating flow

ANCHOR: Morning theory → lunch → farm tour → kit practice → aquaponics theory → assessment.

- Farm tour and kit practice use checklist PDFs
- 1 hour assessment: WA-SAQ then PP

1

Farm · tour · practice · checklist · PDFs

2

hour · assessment · WA-SAQ · then

Assessment evidence

ANCHOR: Competence requires both knowledge and performance evidence.

- WA-SAQ: 4 open-ended questions, 30 min
- PP: one practical task covering A1–A4, 30 min

1

WA-SAQ · open-ended · questions

2

practical · task · covering · A1–A4

Safety stop conditions

ANCHOR: Stop and notify the trainer when a condition is unsafe or uncontrolled.

- Damaged electrical equipment or water near unprotected power
- Unknown chemical, missing label/SDS/PPE, unstable structure

1 Damaged · electrical · equipment · water · near · unprotected

2 Unknown · chemical · missing · label/SDS/PPE · unstable · structure

Measurement conventions

ANCHOR: Record value, unit, time, instrument and action.

- pH is unitless; EC uses mS/cm or μ S/cm
- Temperature and calibration status support interpretation

1 unitless · uses · mS/cm · μ S/cm

2 Temperature · calibration · status · support · interpretation

01

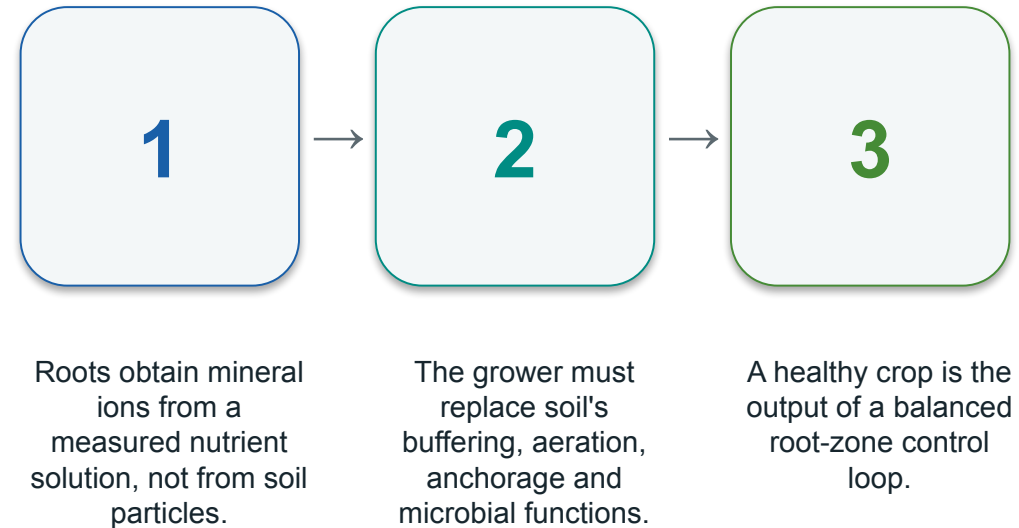
Hydroponic foundations

How water, oxygen, ions and plant demand interact

Hydroponics is a controlled root-zone system

ANCHOR: water + oxygen + ions + support

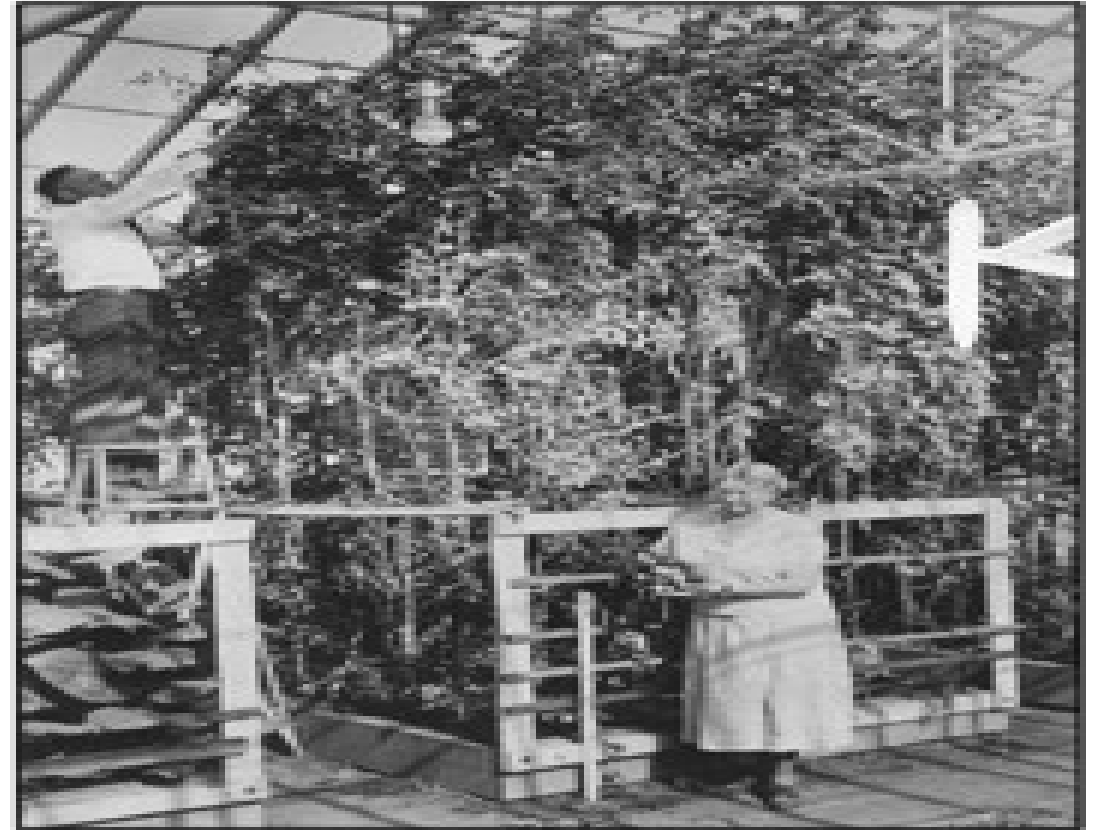
- Roots obtain mineral ions from a measured nutrient solution, not from soil particles.
- The grower must replace soil's buffering, aeration, anchorage and microbial functions.
- A healthy crop is the output of a balanced root-zone control loop.



The four root-zone variables

ANCHOR: EC · pH · DO · temperature

- EC estimates total dissolved ionic strength; it does not identify each element.
- pH governs nutrient availability and chemical stability.
- Dissolved oxygen and solution temperature shape respiration and root disease risk.



Legacy visual reused from original slide 22

Mass balance explains reservoir drift

ANCHOR: inputs – plant uptake – discharge = change

- Water loss through transpiration can raise EC even when no fertiliser is added.
- Selective ion uptake means a stable EC can still hide nutrient imbalance.
- Top-up, correction and periodic replacement are different control actions.

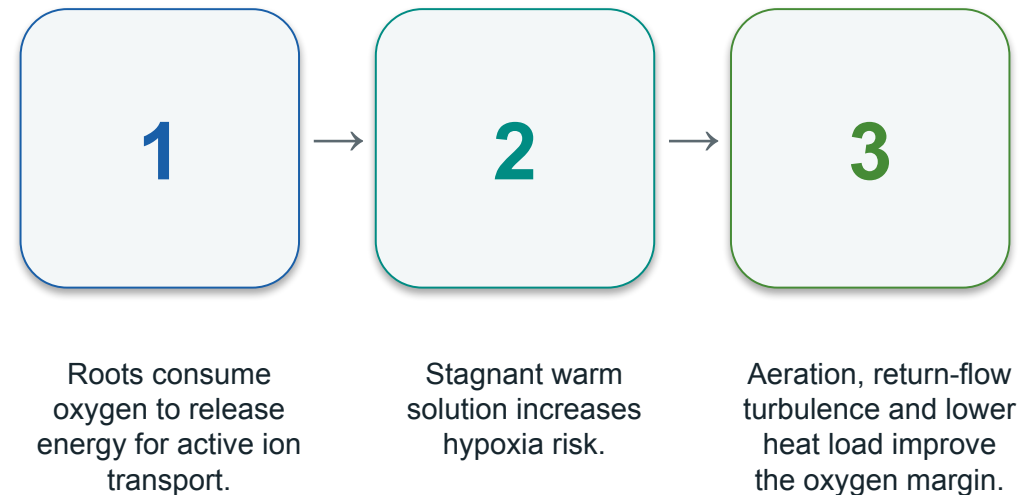


Legacy visual reused from original slide 24

Root respiration sets an oxygen constraint

ANCHOR: warm water holds less dissolved O₂

- Roots consume oxygen to release energy for active ion transport.
- Stagnant warm solution increases hypoxia risk.
- Aeration, return-flow turbulence and lower heat load improve the oxygen margin.



Capillary and film flow are not interchangeable

ANCHOR: water depth changes air-filled porosity

- A thin nutrient film preserves a moist, oxygen-rich root surface.
- Deep flooding provides thermal mass but needs active aeration.
- Media pores can wick water while maintaining structural support.

1 thin · nutrient · film · preserves · moist · oxygen-rich

2 Deep · flooding · provides · thermal · mass · needs

3 Media · pores · wick · water · while · maintaining

Plant water moves down a potential gradient

ANCHOR: root uptake → xylem → stomata

- Light, heat, humidity and airflow change transpiration demand.
- Roots must supply water fast enough to maintain cell turgor.
- Wilting can occur despite a full reservoir when roots are hypoxic or diseased.



Legacy visual reused from original slide 46

Nutrient ions are not plant food

ANCHOR: photosynthesis supplies carbon energy

- Plants build carbohydrates from CO_2 and water using light energy.
- Mineral ions enable enzymes, structure and osmotic regulation.
- More fertiliser cannot compensate for inadequate light or damaged roots.



Legacy visual reused from original slide 48

Soil-free does not mean risk-free

ANCHOR: one reservoir can connect every plant

- Recirculation conserves water but can spread a waterborne pathogen rapidly.
- Shared chemistry means one dosing error affects the whole crop.
- Monitoring and sanitation are system controls, not optional housekeeping.

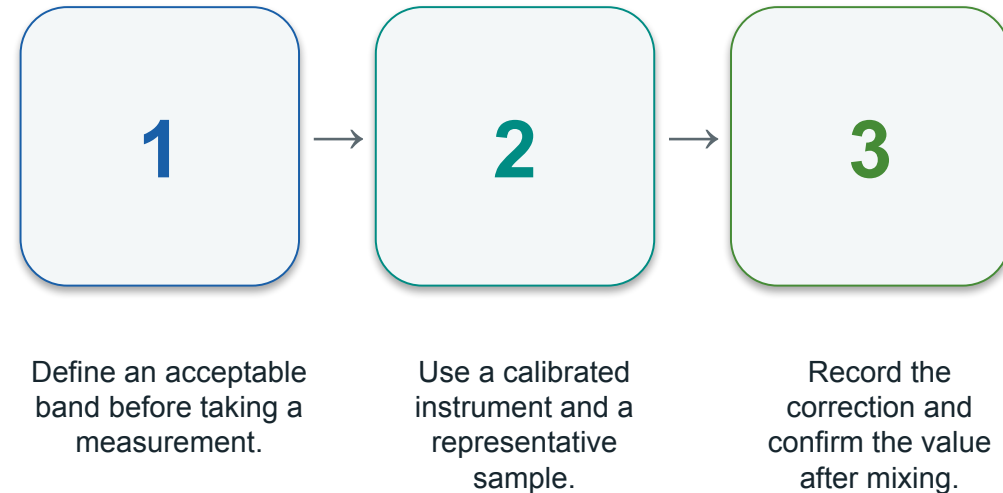


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The production loop is observable

ANCHOR: setpoint → measure → compare → act → verify

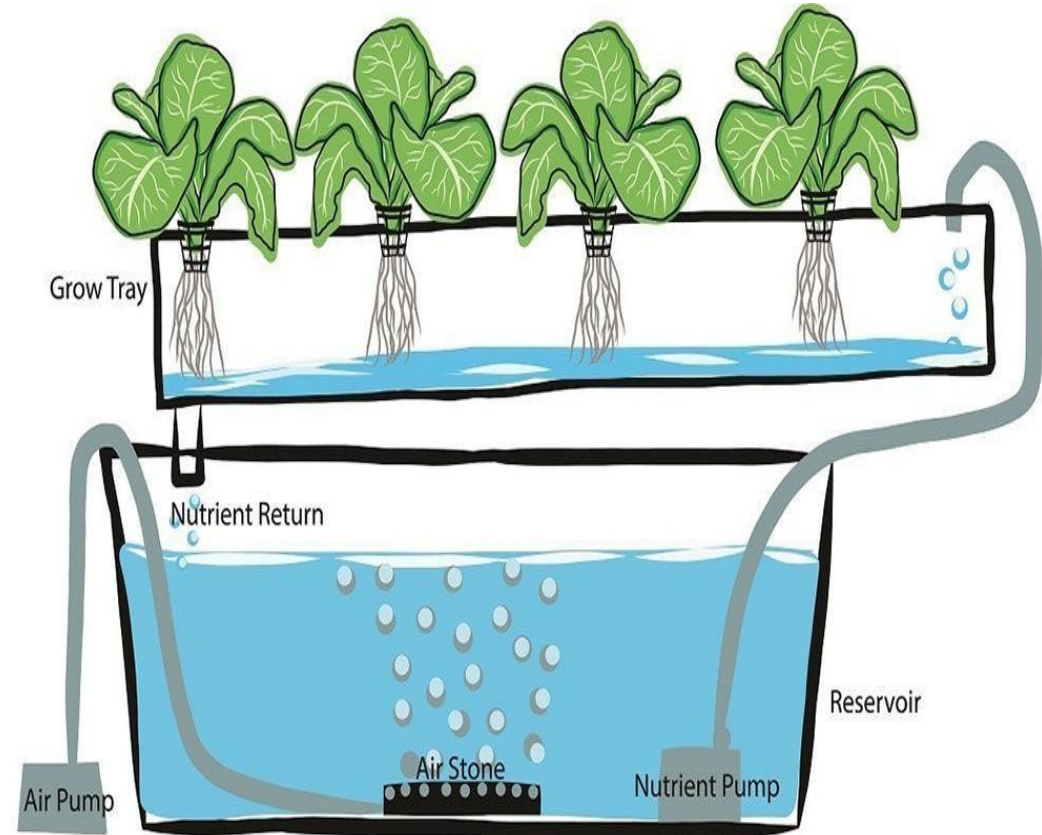
- Define an acceptable band before taking a measurement.
- Use a calibrated instrument and a representative sample.
- Record the correction and confirm the value after mixing.



Evidence distinguishes control from guesswork

ANCHOR: time-stamped log + calibration + crop response

- A single reading without time, unit and instrument has weak evidential value.
- Trend data reveals drift before visible symptoms appear.
- Crop photographs should show both canopy and roots, not leaves alone.



Legacy visual reused from original slide 66

02

Hydroponic systems

Select and diagnose flow, support and failure tolerance

Nutrient Film Technique (NFT)

ANCHOR: shallow recirculating film

- Channel slope and flow continuity are critical; loss of pumping exposes roots quickly.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 75

Deep Water Culture (DWC)

ANCHOR: roots suspended in aerated solution

- Air delivery and raft hygiene govern root oxygen and disease risk.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 80

Kratky passive culture

ANCHOR: static reservoir + deliberate air gap

- A growing air gap supplies oxygen; refilling to the original brim can drown roots.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 87

Ebb-and-flow

ANCHOR: timed flood and drain

- Flood duration wets media; complete drainage restores air-filled porosity.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 91

Drip hydroponics

ANCHOR: emitter uniformity + drainage fraction

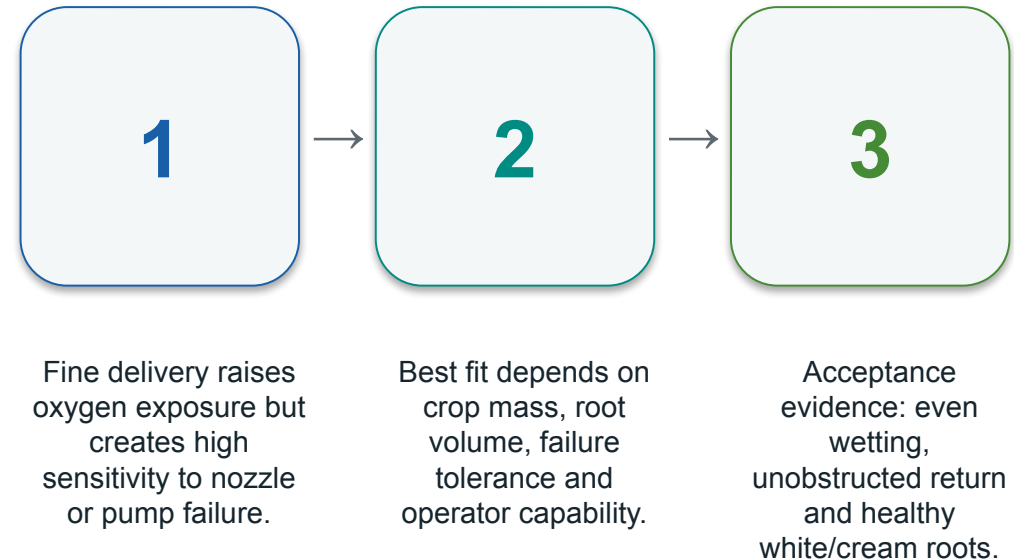
- Emitter blockage and uneven pressure create plant-to-plant variability.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.

**emitter uniformity +
drainage fraction**

Aeroponics

ANCHOR: mist droplet delivery to bare roots

- Fine delivery raises oxygen exposure but creates high sensitivity to nozzle or pump failure.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Wick systems

ANCHOR: passive capillary transport

- Low complexity suits small plants; transport rate limits high-demand crops.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.



Legacy visual reused from original slide 110

Dutch bucket

ANCHOR: individual media containers + common return

- Isolation aids crop support, while a common return retains shared-solution risk.
- Best fit depends on crop mass, root volume, failure tolerance and operator capability.
- Acceptance evidence: even wetting, unobstructed return and healthy white/cream roots.

**individual media
containers + common
return**

System selection is a constraint decision

ANCHOR: crop × site × utilities × operator

- Fast leafy greens tolerate different support and cycle times from fruiting crops.
- A rooftop site adds heat, wind, loading and access constraints.
- The simplest system that reliably meets the production target is often the best choice.

1 Fast · leafy · greens · tolerate · different · support

2 rooftop · site · adds · heat · wind · loading

3 simplest · system · that · reliably · meets · production

Open and recirculating loops trade risk

ANCHOR: discharge loss versus shared-reservoir exposure

- Open drain-to-waste operation simplifies chemistry but loses water and nutrients.
- Recirculation improves resource efficiency and demands stronger monitoring and hygiene.
- Local discharge rules and nutrient recovery should be considered in the SOP.

1 Open · drain-to-waste · operation · simplifies · chemistry · loses

2 Recirculation · improves · resource · efficiency · demands · stronger

3 Local · discharge · rules · nutrient · recovery · should

Pump duty must include static and friction head

ANCHOR: duty point = flow at total dynamic head

- A pump's maximum flow is measured near zero head and is not the operating flow.
- Long narrow tubing, elbows and elevation add resistance.
- Measure delivered channel flow instead of relying on the box label.

1 pump's · maximum · flow · measured · near · zero

2 Long · narrow · tubing · elbows · elevation · resistance

3 Measure · delivered · channel · flow · instead · relying

Return design protects against overflow

ANCHOR: inlet flow $\leq$ unobstructed drain capacity

- Roots, media and biofilm reduce drainage capacity over time.
- A freeboard and emergency overflow limit water damage.
- Power-off testing reveals siphon and reservoir-capacity hazards.

1 **Roots · media · biofilm · reduce · drainage · capacity**

2 **freeboard · emergency · overflow · limit · water · damage**

3 **Power-off · testing · reveals · siphon · reservoir-capacity · hazards**

Redundancy is proportional to crop loss exposure

ANCHOR: failure detection time × root drying rate

- NFT may require rapid alarm response because exposed roots dry quickly.
- DWC carries more water reserve but depends on aeration.
- Spare pumps, independent air and tested response actions reduce consequence.

1 require · rapid · alarm · response · because · exposed

2 carries · more · water · reserve · depends · aeration

3 Spare · pumps · independent · tested · response · actions

Media properties drive irrigation strategy

ANCHOR: air-filled porosity · water holding · stability

- Coco coir holds water but requires controlled salt management.
- Perlite increases aeration with low mass and low buffering.
- Rockwool is uniform but demands careful wetness control and disposal planning.



Legacy visual reused from original slide 121

Food-contact materials need verification

ANCHOR: compatible plastics + cleanable geometry

- Avoid unknown reused containers that may have held chemicals.
- Opaque reservoirs suppress light-driven algae growth.
- Smooth accessible surfaces make cleaning and inspection more reliable.

1 Avoid · unknown · reused · containers · that · have

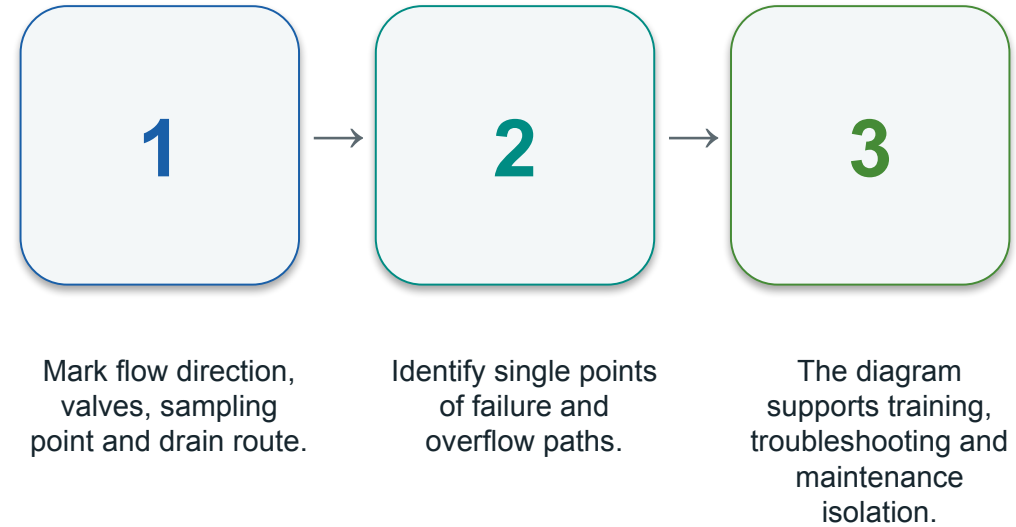
2 Opaque · reservoirs · suppress · light-driven · algae · growth

3 Smooth · accessible · surfaces · make · cleaning · inspection

A system diagram is an operating control

ANCHOR: reservoir → pump → crop → return

- Mark flow direction, valves, sampling point and drain route.
- Identify single points of failure and overflow paths.
- The diagram supports training, troubleshooting and maintenance isolation.



03

Water and nutrients

Measure, mix and correct the root-zone chemistry

Source water sets the starting chemistry

ANCHOR: alkalinity differs from pH

- pH is the current hydrogen-ion condition; alkalinity resists pH change.
- High bicarbonate water can drive pH upward after correction.
- Record source EC, pH and where relevant alkalinity before formulating.

1 current · hydrogen-ion · condition · alkalinity · resists · change

2 High · bicarbonate · water · drive · upward · after

3 Record · source · where · relevant · alkalinity · before

pH controls chemical availability

ANCHOR: leafy-green working band commonly ~5.5–6.5

- Outside the operating band, some ions become less available or precipitate.
- Adjust in small increments after nutrients are fully mixed.
- Crop-specific SOPs override a generic band.

1 Outside · operating · band · some · ions · become

2 Adjust · small · increments · after · nutrients · fully

3 Crop-specific · SOPs · override · generic · band

EC is a proxy, not a recipe

ANCHOR: mS/cm or μ S/cm at a stated temperature

- EC increases with dissolved ions from fertiliser and source water.
- Different ion mixtures can produce a similar EC.
- Use the supplier formulation and crop response together with EC.

1 increases · with · dissolved · ions · from · fertiliser

2 Different · mixtures · produce · similar

3 supplier · formulation · crop · response · together · with

Temperature compensation has limits

ANCHOR: conductivity changes with temperature

- ATC helps standardise meter readings but does not protect warm roots.
- Record solution temperature with pH and EC.
- Compare trends at similar times of day where possible.

1 helps · standardise · meter · readings · does · protect

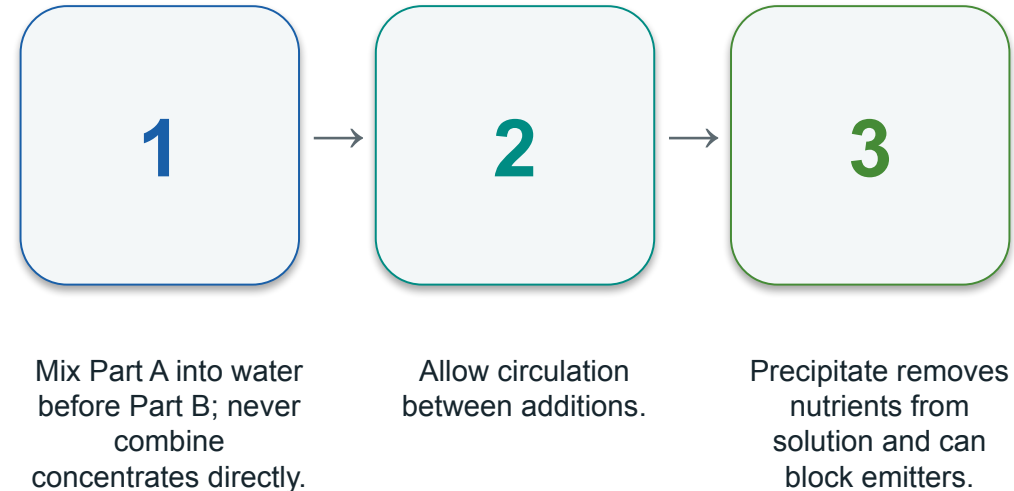
2 Record · solution · temperature · with

3 Compare · trends · similar · times · where · possible

Two-part nutrients prevent concentrate reactions

ANCHOR: keep calcium separate from concentrated sulfate/phosphate

- Mix Part A into water before Part B; never combine concentrates directly.
- Allow circulation between additions.
- Precipitate removes nutrients from solution and can block emitters.



Macronutrients build structure and metabolism

ANCHOR: N · P · K · Ca · Mg · S

- Nitrogen supports proteins and chlorophyll; imbalance can drive weak lush growth.
- Potassium regulates water balance and enzyme activity.
- Calcium depends strongly on transport to new tissue.

1 Nitrogen · supports · proteins · chlorophyll · imbalance · drive

2 Potassium · regulates · water · balance · enzyme · activity

3 Calcium · depends · strongly · transport · tissue

Micronutrients operate at small concentrations

ANCHOR: Fe · Mn · B · Zn · Cu · Mo · Cl · Ni

- Trace requirement does not mean optional.
- Excess micronutrients can become toxic quickly.
- pH and chelation influence iron availability.

1 Trace · requirement · does · mean · optional

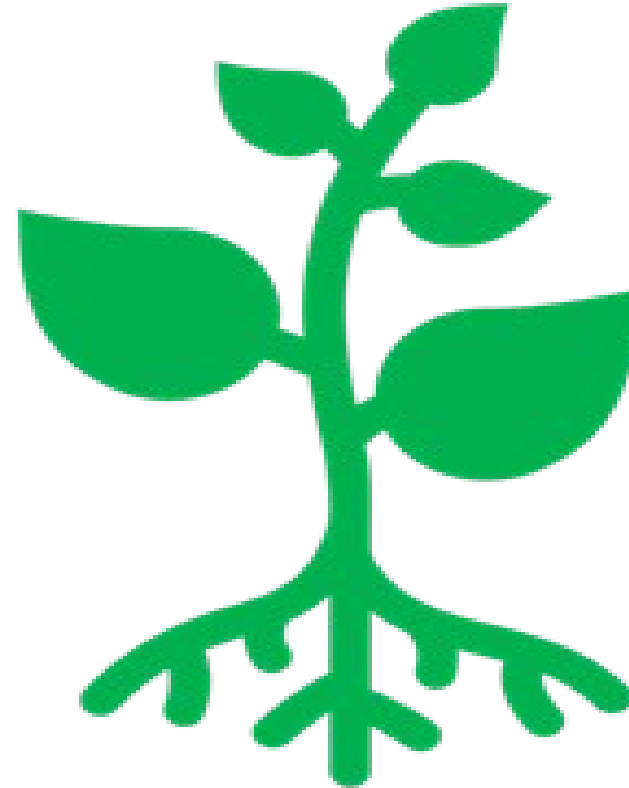
2 Excess · micronutrients · become · toxic · quickly

3 chelation · influence · iron · availability

Symptoms are patterns, not diagnoses

ANCHOR: old versus new growth + distribution + roots

- Mobile-element shortages often appear first on older leaves.
- Immobile-element disorders often affect new growth.
- Confirm root health, pH, EC and recent operations before correcting nutrients.



Legacy visual reused from original slide 133

Tip burn can occur with calcium present

ANCHOR: calcium transport follows transpiration

- Fast inner-leaf growth, high humidity and weak airflow can limit local calcium delivery.
- Raising EC blindly may worsen water uptake.
- Control growth rate, air movement and root conditions.

1 Fast · inner-leaf · growth · high · humidity · weak

2 Raising · blindly · worsen · water · uptake

3 Control · growth · rate · movement · root · conditions

Iron chlorosis needs differential diagnosis

ANCHOR: interveinal yellowing on young leaves

- High root-zone pH can reduce iron availability.
- Root damage and cold/wet stress can mimic nutrient shortage.
- Correct the limiting condition before adding more concentrate.

1 High · root-zone · reduce · iron · availability

2 Root · damage · cold/wet · stress · mimic · nutrient

3 Correct · limiting · condition · before · adding · more

Dilution calculation prevents overdosing

ANCHOR: $C_1 V_1 = C_2 V_2$

- Use consistent units and confirm the concentrate label.
- Estimate conservatively, mix, measure, then trim.
- Wear required PPE and keep acids/bases labelled and segregated.

1 consistent · units · confirm · concentrate · label

2 Estimate · conservatively · measure · then · trim

3 Wear · required · keep · acids/bases · labelled · segregated

Top-up strategy depends on both water and ions

ANCHOR: reservoir level + EC trend + crop stage

- Falling level with rising EC suggests water is leaving faster than ions.
- Falling level with falling EC suggests strong nutrient uptake or dilution history.
- Periodic replacement resets selective-ion drift and accumulated residues.

1 Falling · level · with · rising · suggests · water

2 Falling · level · with · falling · suggests · strong

3 Periodic · replacement · resets · selective-ion · drift · accumulated

Meter calibration is part of the measurement

ANCHOR: known standard → rinse → calibrate → verify

- Use fresh, unexpired standards appropriate to the meter.
- Do not pour used standard back into its bottle.
- Record calibration date, standard value and result.

1 fresh · unexpired · standards · appropriate · meter

2 pour · used · standard · back · into · bottle

3 Record · calibration · date · standard · value · result

Sampling location changes the reading

ANCHOR: representative, mixed, bubble-free sample

- A dosing point can read very differently before complete mixing.
- Rinse the cup and probe with sample solution.
- Consistent sampling improves trend comparability.

1 dosing · point · read · very · differently · before

2 Rinse · probe · with · sample · solution

3 Consistent · sampling · improves · trend · comparability

pH correction is a controlled chemical task

ANCHOR: small dose + circulation + remeasurement

- Concentrated acid or alkali can burn skin and roots.
- Add chemical to water as specified; never improvise mixing order.
- Stop if the label, SDS, PPE or correct dosing tool is unavailable.

1 Concentrated · acid · alkali · burn · skin · roots

2 chemical · water · specified · never · improvise · mixing

3 Stop · label · correct · dosing · tool · unavailable

Reservoir sizing buffers change

ANCHOR: litres per plant and refill frequency

- A larger volume changes temperature and EC more slowly.
- Excess volume adds weight and cleaning burden.
- Keep enough free capacity for return water during power-off tests.

1 larger · volume · changes · temperature · more · slowly

2 Excess · volume · adds · weight · cleaning · burden

3 Keep · enough · free · capacity · return · water

Light exclusion is an algae control

ANCHOR: photons + nutrients + wet surface = algae opportunity

- Cover reservoirs and unused planting holes.
- Algae competes for oxygen, blocks flow and creates slippery surfaces.
- Clean the root cause rather than relying only on chemical treatment.

1

Cover · reservoirs · unused · planting · holes

2

Algae · competes · oxygen · blocks · flow · creates

3

Clean · root · cause · rather · than · relying

Biofilm changes hydraulics and hygiene

ANCHOR: surface attachment + organic load + time

- Biofilm can harbour organisms and narrow emitters or channels.
- Inspection targets junctions, dead legs and rough surfaces.
- Cleaning must include physical removal, validated chemistry and rinsing.

1 Biofilm · harbour · organisms · narrow · emitters · channels

2 Inspection · targets · junctions · dead · legs · rough

3 Cleaning · must · include · physical · removal · validated

04

Crop production

Control nursery, environment, hygiene and crop health

Crop choice must fit the system envelope

ANCHOR: cycle time · canopy · root volume · support

- Leafy greens suit short cycles and modest structural support.
- Fruiting crops need longer occupancy, pollination and trellising.
- Marketable yield, not plant count, is the production measure.

1 Leafy · greens · suit · short · cycles · modest

2 Fruiting · crops · need · longer · occupancy · pollination

3 Marketable · yield · plant · count · production · measure

Seed quality affects the whole batch

ANCHOR: germination % and uniform emergence

- Record cultivar, lot and sowing date.
- Uniform plugs create more even transplant and harvest timing.
- Separate non-germination from later seedling mortality.

1 Record · cultivar · sowing · date

2 Uniform · plugs · create · more · even · transplant

3 Separate · non-germination · from · later · seedling · mortality

Nursery moisture balances water and oxygen

ANCHOR: saturated is not synonymous with healthy

- Overwet media reduces aeration and encourages disease.
- Drying interrupts germination and damages fine roots.
- Use clean water delivery and defined transplant acceptance criteria.

1 Overwet · media · reduces · aeration · encourages · disease

2 Drying · interrupts · germination · damages · fine · roots

3 clean · water · delivery · defined · transplant · acceptance

Transplant timing is a root-and-canopy decision

ANCHOR: established roots + true leaves + intact plug

- Overgrown seedlings suffer root binding and transition stress.
- Weak seedlings create permanent variability in channel performance.
- Reject diseased or mechanically damaged plugs.

1 Overgrown · seedlings · suffer · root · binding · transition

2 Weak · seedlings · create · permanent · variability · channel

3 Reject · diseased · mechanically · damaged · plugs

Light quantity is measured as photons

ANCHOR: PPFD ($\mu\text{mol}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$) and DLI

- PPFD describes instantaneous photosynthetic photon flux density.
- DLI integrates PPFD over the photoperiod.
- Uniformity matters because the lowest-light positions limit consistent production.

1 PPFD · describes · instantaneous · photosynthetic · photon · flux

2 integrates · PPFD · over · photoperiod

3 Uniformity · matters · because · lowest-light · positions · limit

Daily Light Integral links intensity and time

ANCHOR: $DLI = PPFD \times \text{seconds} \div 1,000,000$

- Longer photoperiod can partly compensate for lower intensity within crop limits.
- More photons can raise cooling and electricity demand.
- Measure at canopy height across the growing area.

1 Longer · photoperiod · partly · compensate · lower · intensity

2 More · photons · raise · cooling · electricity · demand

3 Measure · canopy · height · across · growing · area

Air temperature and humidity shape demand

ANCHOR: vapour-pressure deficit (VPD)

- Very low evaporative demand can reduce nutrient transport to fast-growing tissue.
- Very high demand can exceed root water supply and close stomata.
- Use crop response with environmental trends, not temperature alone.

1 Very · evaporative · demand · reduce · nutrient · transport

2 Very · high · demand · exceed · root · water

3 crop · response · with · environmental · trends · temperature

Airflow is a boundary-layer control

ANCHOR: leaf movement without mechanical stress

- Gentle circulation supports gas exchange and canopy uniformity.
- Dead zones retain humidity and favour disease.
- Excess wind raises water demand and causes abrasion.

1 Gentle · circulation · supports · exchange · canopy · uniformity

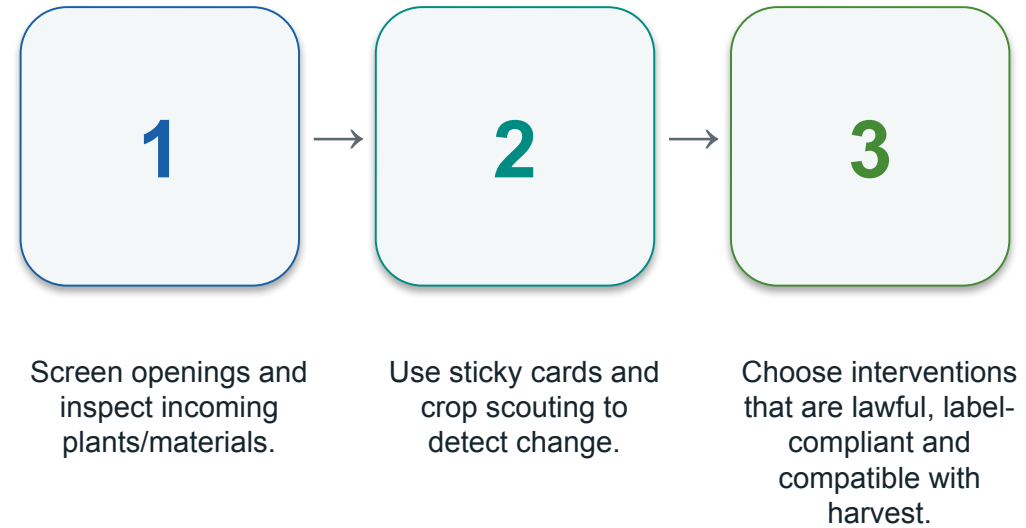
2 Dead · zones · retain · humidity · favour · disease

3 Excess · wind · raises · water · demand · causes

Integrated pest management starts with exclusion

ANCHOR: prevent → monitor → identify → intervene → verify

- Screen openings and inspect incoming plants/materials.
- Use sticky cards and crop scouting to detect change.
- Choose interventions that are lawful, label-compliant and compatible with harvest.



Sticky cards are trend tools

ANCHOR: standard location + interval + count

- Card colour and placement affect which flying pests are captured.
- Counts are comparable only with a consistent method.
- A trap result must be paired with crop inspection and correct identification.

1 Card · colour · placement · affect · which · flying

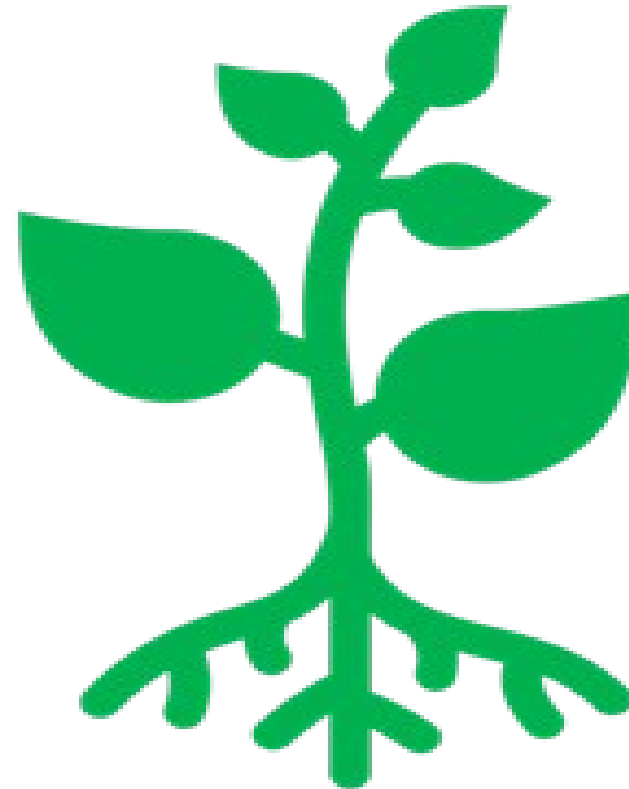
2 Counts · comparable · only · with · consistent · method

3 trap · result · must · paired · with · crop

Root inspection is non-negotiable

ANCHOR: colour · odour · firmness · branching

- Healthy roots are generally light-coloured and firm, with crop-dependent variation.
- Brown slimy roots and off-odour require rapid investigation.
- Inspect the least vigorous plants and the return end, not only the best canopy.



Legacy visual reused from original slide 133

Harvest hygiene protects ready-to-eat produce

ANCHOR: clean hands/tools + cleanable surfaces + lot traceability

- Separate dirty maintenance tools from harvest equipment.
- Remove damaged or contaminated produce from the marketable lot.
- Record harvest date, crop/lot, quantity and responsible operator.

1 Separate · dirty · maintenance · tools · from · harvest

2 Remove · damaged · contaminated · produce · from · marketable

3 Record · harvest · date · crop/lot · quantity · responsible

Housekeeping controls slips and contamination

ANCHOR: dry walkways + contained chemicals + clear access

- Leaks are production deviations and safety hazards.
- Electrical connections need dry, protected routing.
- Emergency access, labels and SDS must remain visible.

1 Leaks · production · deviations · safety · hazards

2 Electrical · connections · need · protected · routing

3 Emergency · access · labels · must · remain · visible

Cleaning is not the same as disinfection

ANCHOR: soil removal precedes disinfectant contact

- Organic residue shields organisms and consumes disinfectant.
- Correct concentration, contact time and material compatibility are required.
- Rinsing and drying requirements follow the product label/SOP.

1 Organic · residue · shields · organisms · consumes · disinfectant

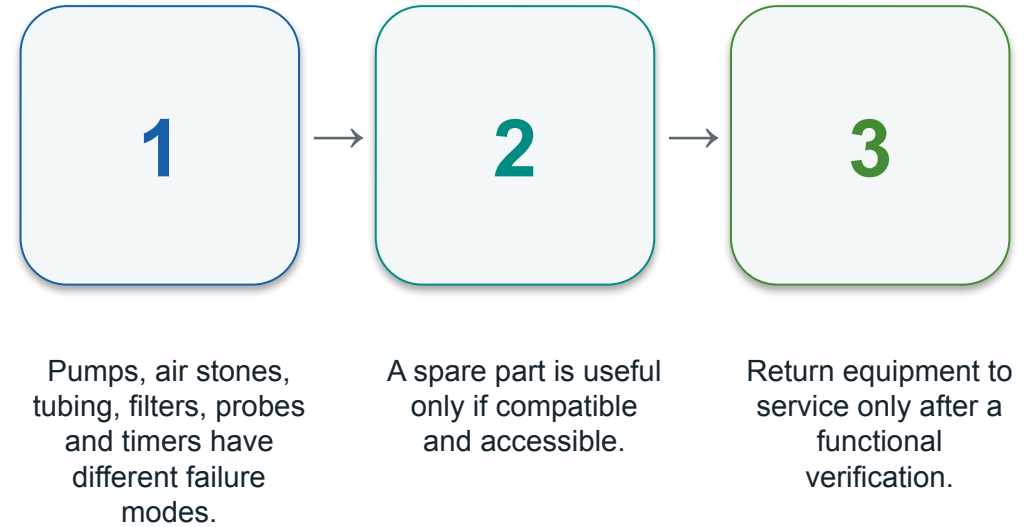
2 Correct · concentration · contact · time · material · compatibility

3 Rinsing · drying · requirements · follow · product · label/SOP

Maintenance preserves control capability

ANCHOR: inspect → clean → replace → function-test → record

- Pumps, air stones, tubing, filters, probes and timers have different failure modes.
- A spare part is useful only if compatible and accessible.
- Return equipment to service only after a functional verification.



05

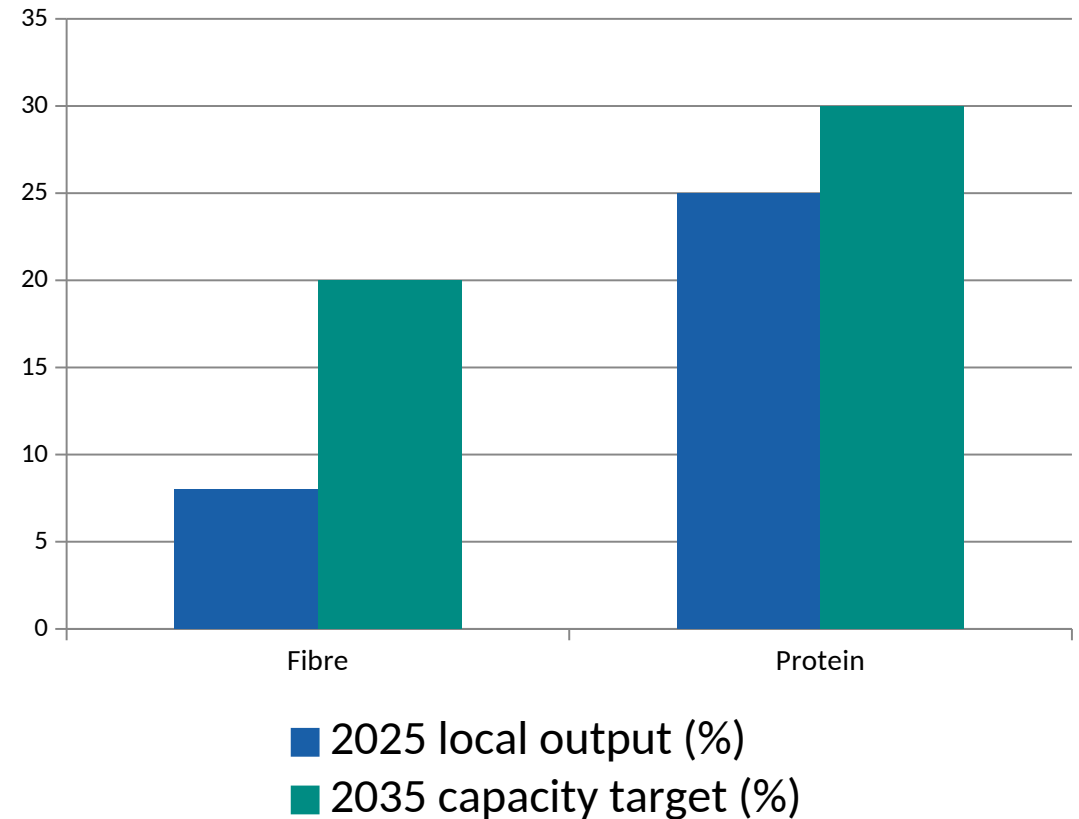
Singapore hydroponics

Current policy, farms, productivity and technology cases

Singapore Food Story 2 resets the context

ANCHOR: 2035 capacity targets—20% fibre, 30% protein

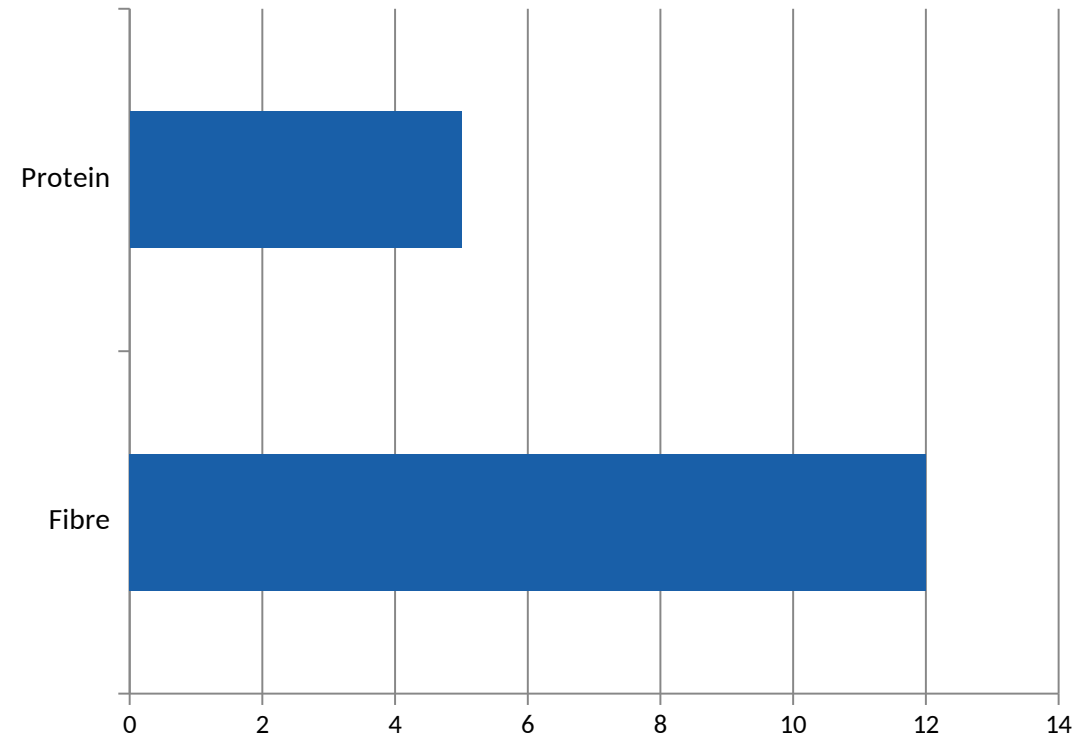
- The current policy focuses on resilient production capacity and commercial viability.
- Previous national framing has been superseded by the current 2035 capacity targets.
- Hydroponics contributes mainly to the fibre category through vegetables.



2025 output shows the remaining fibre gap

ANCHOR: about 8% fibre and 25% protein in 2025

- Local output is only one layer of Singapore's food-resilience strategy.
- Productivity, demand, import diversification and stockpiling interact.
- A technical system must still produce saleable crops at viable cost.

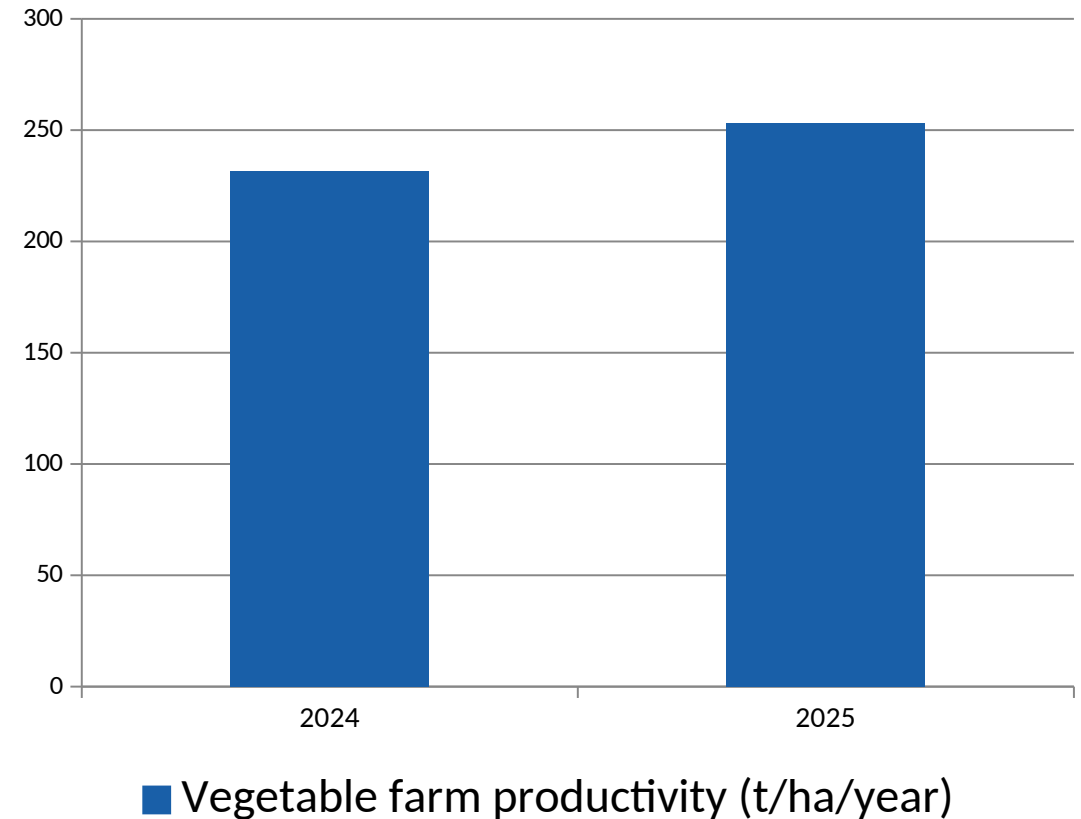


■ Remaining percentage-point gap to 2035 target

Vegetable productivity rose in 2025

ANCHOR: 231.4 → 253.3 tonnes/ha/year

- SFA reported roughly a 10% increase in farm productivity.
- Area-normalised output supports comparison but does not reveal energy or cost.
- Pair yield with grade-out, input use, labour and downtime.



ComCrop demonstrates rooftop hybrid production

ANCHOR: rooftop hydroponics + sunlight + supplemental light

- SFA describes ComCrop as Singapore's first commercial rooftop farm, established in 2011.
- Sensors support monitoring of water, light and nutrient conditions.
- Manual transplanting, harvesting and cleaning remain important operations.



Legacy visual reused from original slide 46

Sustenir compresses production into controlled space

ANCHOR: NFT + vertical layers + controlled environment

- SFA describes recirculating NFT with temperature, humidity and CO₂ control.
- The farm page reports 3.2 tonnes of lettuce from 54 m² and over 90 tonnes/year.
- Treat site figures as reported case data, not universal performance.

1 describes · recirculating · with · temperature · humidity · control

2 farm · page · reports · tonnes · lettuce · from

3 Treat · site · figures · reported · case · data

Singrow targets temperate crops in the tropics

ANCHOR: cultivar + climate control + automation

- The NUS Enterprise case describes proprietary hydroponic strawberries.
- Computer vision, robotics and crop genetics form one production system.
- Higher-value crops may justify complexity that leafy greens cannot.

1 Enterprise · case · describes · proprietary · hydroponic · strawberries

2 Computer · vision · robotics · crop · genetics · form

3 Higher-value · crops · justify · complexity · that · leafy

Greenphyto signals a scale-and-automation bet

ANCHOR: reported 23 m chambers and up to 2,000 t/year

- The Business Times reported an S\$80 million multi-storey hydroponic facility in January 2026.
- The report describes automated seeding, dosing, crop movement and climate control.
- Capacity and company claims must be distinguished from verified realised output.

1 Business · Times · reported · S\$80 · million · multi-storey

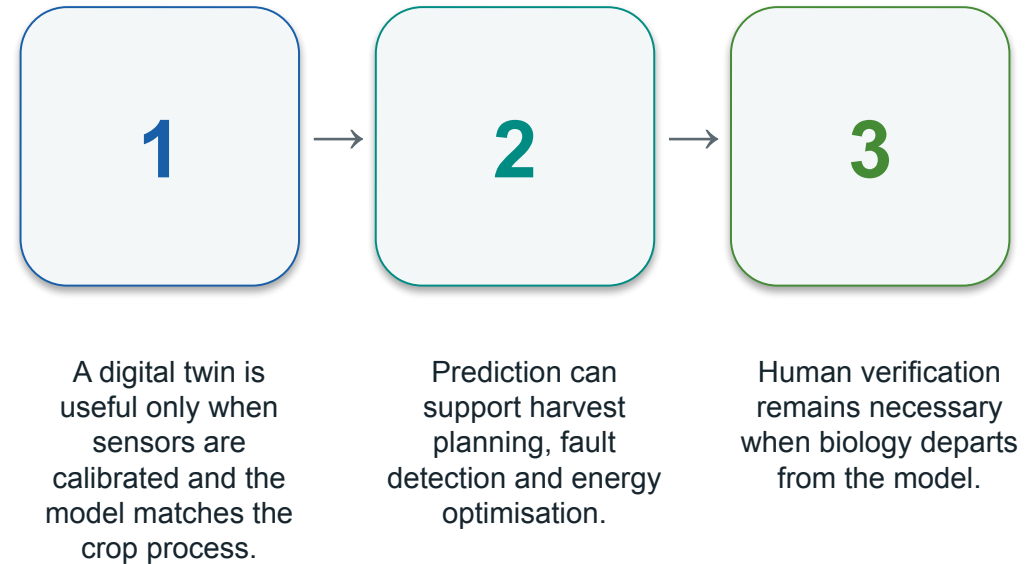
2 report · describes · automated · seeding · dosing · crop

3 Capacity · company · claims · must · distinguished · from

Digital twins turn operations into models

ANCHOR: sensor state → model → prediction → action

- A digital twin is useful only when sensors are calibrated and the model matches the crop process.
- Prediction can support harvest planning, fault detection and energy optimisation.
- Human verification remains necessary when biology departs from the model.



Automation changes the job, not the biology

ANCHOR: actuators still depend on trustworthy measurements

- Automated dosing magnifies a bad probe or wrong setpoint.
- Alarms require ownership, escalation and tested response time.
- Operators need crop observation plus technical troubleshooting skills.

1 Automated · dosing · magnifies · probe · wrong · setpoint

2 Alarms · require · ownership · escalation · tested · response

3 Operators · need · crop · observation · plus · technical

ACT Fund 2 supports capability transformation

ANCHOR: new S\$70m tranche from 2026

- SFA lists capability upgrading, technology adoption and standards pathways.
- A grant does not remove the need for business and technical feasibility.
- Define measurable production and capability outcomes before procurement.

1 lists · capability · upgrading · technology · adoption · standards

2 grant · does · remove · need · business · technical

3 Define · measurable · production · capability · outcomes · before

SG GAP connects production with assurance

ANCHOR: food safety + quality + environmental + worker welfare controls

- Good agricultural practice depends on documented, repeatable controls.
- Records support traceability and corrective action.
- Certification scope and current requirements must be checked with SFA/standards sources.

1 Good · agricultural · practice · depends · documented · repeatable

2 Records · support · traceability · corrective · action

3 Certification · scope · current · requirements · must · checked

Land productivity alone can mislead

ANCHOR: yield/m² must be read with kWh, labour, water and grade-out

- Vertical stacking can raise output per floor area.
- Lighting, cooling, pumping and dehumidification add operating load.
- A resilient farm optimises the whole system, not a single headline metric.

1

Vertical · stacking · raise · output · floor · area

2

Lighting · cooling · pumping ·
dehumidification · operating · load

3

resilient · farm · optimises · whole · system ·
single

Tropical heat is a root-zone design input

ANCHOR: heat gain → warmer solution → lower oxygen margin

- Shade, insulation, reservoir location and operating time influence heat load.
- Cooling can protect crops but adds capital and energy cost.
- Measure the actual daily temperature profile before choosing controls.

1 Shade · insulation · reservoir · location · operating · time

2 Cooling · protect · crops · adds · capital · energy

3 Measure · actual · daily · temperature · profile · before

Farm economics start with marketable yield

ANCHOR: revenue = saleable mass × realised price

- High biological yield can coexist with poor grade-out or weak demand.
- Cycle time and space occupancy affect annual throughput.
- Include packaging, logistics, rejects, energy, labour and downtime.

1 High · biological · yield · coexist · with · poor

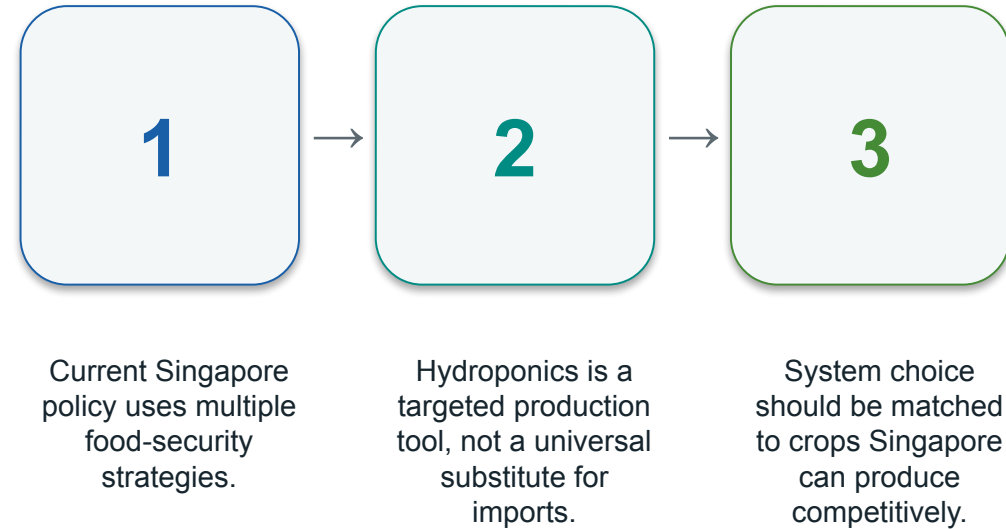
2 Cycle · time · space · occupancy · affect · annual

3 Include · packaging · logistics · rejects · energy · labour

Resilience needs more than local farms

ANCHOR: local capacity + diversified imports + stockpiles + partnerships

- Current Singapore policy uses multiple food-security strategies.
- Hydroponics is a targeted production tool, not a universal substitute for imports.
- System choice should be matched to crops Singapore can produce competitively.



06

Farm tour activity

Observe MEOD Farm using evidence-based checklists

Lunch break

Resume at the farm-tour assembly point stated by the trainer

Observe flow before judging appearance

ANCHOR: trace reservoir → delivery → root zone → return

- Identify the system boundary and where water can leave it.
- Locate measurement, dosing, filtration and overflow points.
- Record observations; do not touch equipment without permission.



Visual: generated for this course

Read the crop as a spatial pattern

ANCHOR: inlet/middle/return + upper/lower tier

- Uniform symptoms suggest a shared input; local symptoms suggest position-specific causes.
- Compare canopy colour, size and density across the system.
- Photographs require farm permission and must avoid faces or confidential displays.

1 Uniform · symptoms · suggest · shared · input · local

2 Compare · canopy · colour · size · density · across

3 Photographs · require · farm · permission · must · avoid

Audit water and nutrient controls

ANCHOR: instrument + setpoint + frequency + response

- Ask where and when pH, EC and temperature are measured.
- Look for calibration, dosing and corrective-action records.
- Distinguish observed evidence from an operator explanation.

1 where · when · temperature · measured

2 Look · calibration · dosing · corrective-action · records

3 Distinguish · observed · evidence · from · operator · explanation

Audit WSH and food-safety controls

ANCHOR: hazard → existing control → residual risk

- Observe wet floors, electrical routing, chemical storage and manual handling.
- Look for clean/dirty zoning and harvest-tool control.
- Escalate urgent hazards to the trainer; never intervene in machinery.

1 Observe · floors · electrical · routing · chemical · storage

2 Look · clean/dirty · zoning · harvest-tool · control

3 Escalate · urgent · hazards · trainer · never · intervene

Map preventive maintenance

ANCHOR: asset ID + interval + completion + function test

- Check pumps, filters, air delivery, sensors, lighting and structural supports.
- Ask how failure is detected and who responds.
- Look for spare-part readiness and documented return-to-service tests.

1 Check · pumps · filters · delivery · sensors · lighting

2 failure · detected · responds

3 Look · spare-part · readiness · documented · return-to-service · tests

Evaluate pest management evidence

ANCHOR: monitoring location + count trend + intervention record

- Identify exclusion barriers, scouting routines and sticky-card placement.
- Ask how pests are identified before treatment.
- Confirm interventions respect label, food-safety and pre-harvest requirements.

1 Identify · exclusion · barriers · scouting · routines · sticky-card

2 pests · identified · before · treatment

3 Confirm · interventions · respect · label · food-safety · pre-harvest

Separate claim, observation and inference

ANCHOR: C/O/I notation

- Claim: what the guide/operator says.
- Observation: what you directly see, measure or read.
- Inference: your explanation, labelled as provisional until verified.

1

Claim · what · guide/operator · says

2

Observation · what · directly · measure · read

3

Inference · your · explanation · labelled · provisional · until

Farm-tour evidence is a checklist package

ANCHOR: completed PDF + permitted photos + one flow sketch

- The activity PDF defines the required evidence.
- Use concise factual notes with location and time.
- Submit no personal data, trade secrets or unsafe photographs.

1 activity · defines · required · evidence

2 concise · factual · notes · with · location · time

3 Submit · personal · data · trade · secrets · unsafe

07

Hydroponics kit practice

Commission, verify and return a kit to a controlled state

Kit readiness is an acceptance state

ANCHOR: clean · intact · stable · leak-free · electrically safe

- A visual inspection occurs before water or power is applied.
- Tubing must be secured without kinks and returns must be unobstructed.
- Reject damaged cables, unknown containers or unstable structures.

1 visual · inspection · occurs · before · water · power

2 Tubing · must · secured · without · kinks · returns

3 Reject · damaged · cables · unknown · containers · unstable

Flow verification uses evidence

ANCHOR: every site wet + return stable + no overflow

- Observe the last planting site, not only the first.
- Confirm pump noise and vibration are normal for the unit.
- A timed-volume check can quantify delivered flow where required.

1 Observe · last · planting · site · only · first

2 Confirm · pump · noise · vibration · normal · unit

3 timed-volume · check · quantify · delivered · flow · where

A baseline precedes correction

ANCHOR: source water pH/EC/temperature recorded first

- Baseline data separates source-water contribution from added nutrients.
- Use clean labelled tools to avoid cross-contamination.
- The Learner Guide contains the detailed sequence and safety checks.

1 Baseline · data · separates · source-water · contribution · from

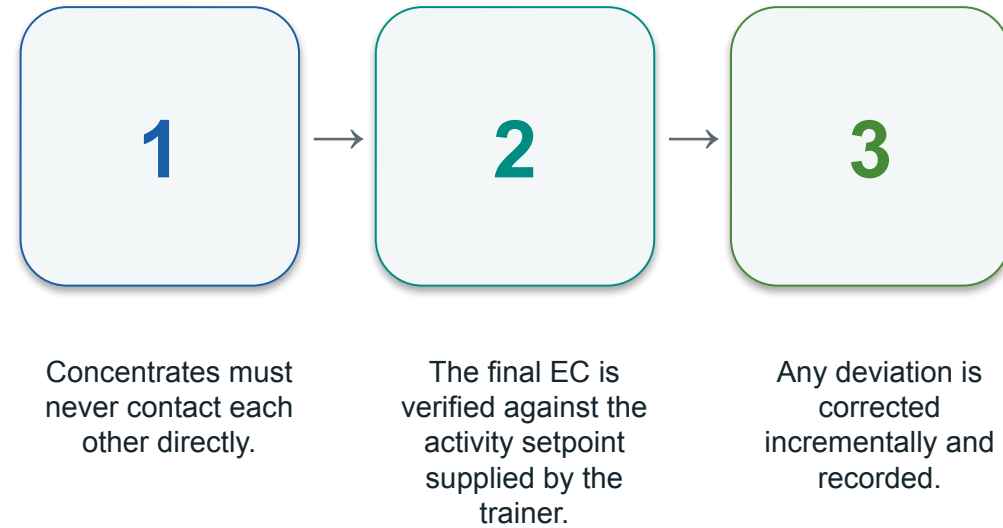
2 clean · labelled · tools · avoid · cross-contamination

3 Learner · Guide · contains · detailed · sequence · safety

Mixing order protects chemistry

ANCHOR: water → A → circulate → B → circulate → measure

- Concentrates must never contact each other directly.
- The final EC is verified against the activity setpoint supplied by the trainer.
- Any deviation is corrected incrementally and recorded.



pH adjustment uses a stop condition

ANCHOR: no label/SDS/PPE = no dosing

- Use the trainer-provided product and dedicated dosing tool only.
- Add small amounts, circulate and remeasure.
- Never return excess chemical to the stock container.

1 **trainer-provided · product · dedicated · dosing
· tool · only**

2 **small · amounts · circulate · remeasure**

3 **Never · return · excess · chemical · stock ·
container**

Transplant acceptance protects uniformity

ANCHOR: intact plug + healthy root + centred support + stable crown

- Roots should contact moisture without burying the crown.
- Remove weak or diseased seedlings from the production batch.
- Plant labels preserve cultivar and date traceability.

1 Roots · should · contact · moisture · without · burying

2 Remove · weak · diseased · seedlings · from · production

3 Plant · labels · preserve · cultivar · date · traceability

Power-off behaviour is part of commissioning

ANCHOR: no siphon overflow + reservoir contains return volume

- Simulated power loss exposes drainage and restart behaviour.
- Restart must restore delivery to every site.
- Do not conduct the test where electrical or spill controls are inadequate.

1 Simulated · power · loss · exposes · drainage · restart

2 Restart · must · restore · delivery · every · site

3 conduct · test · where · electrical · spill · controls

A correction log closes the loop

ANCHOR: before → action → after → operator → time

- Record the reason and exact quantity added.
- Allow adequate mixing before the after-reading.
- Escalate if the value does not respond as expected.

1 Record · reason · exact · quantity · added

2 Allow · adequate · mixing · before · after-
reading

3 Escalate · value · does · respond · expected

Root-zone diagnosis starts below the canopy

ANCHOR: roots + flow + temperature + chemistry

- Leaf symptoms alone can lead to the wrong correction.
- Inspect for obstruction, low aeration, warmth and odour.
- Isolate suspect plants and protect the shared reservoir where required by SOP.

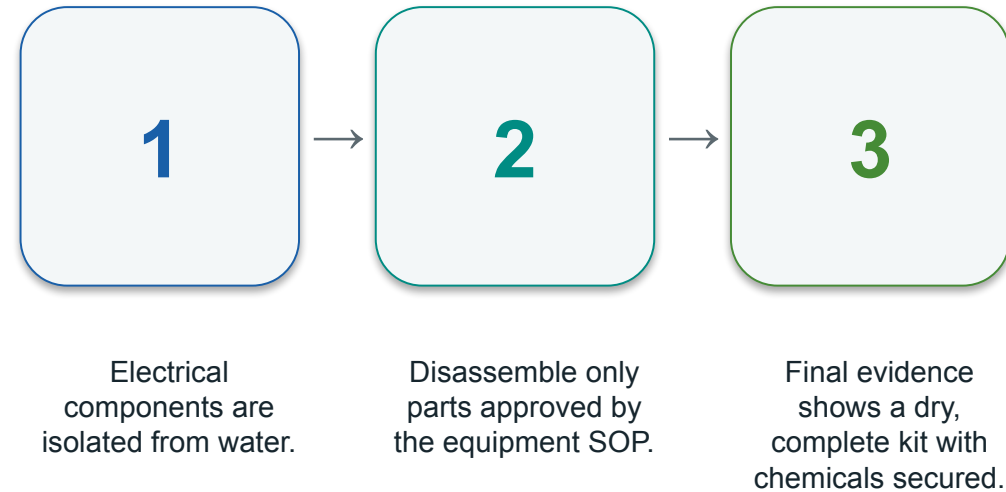


Visual: generated for this course

Cleaning returns the kit to a controlled state

ANCHOR: drain → remove residue → clean → rinse → dry
→ store

- Electrical components are isolated from water.
- Disassemble only parts approved by the equipment SOP.
- Final evidence shows a dry, complete kit with chemicals secured.



Practice evidence maps to A1–A4

ANCHOR: prepare · operate · comply · maintain

- A1: materials and equipment ready.
- A2: system operates to target with measured evidence.
- A3: WSH, food safety and housekeeping controls observed.
- A4: cleaning, maintenance check and storage completed.

1

A1: materials and equipment ready

2

A2: system operates to target with measured evidence

3

A3: WSH, food safety and housekeeping controls observed

4

A4: cleaning, maintenance check and storage completed

08

Aquaponics theory

Balance fish, microbes, plants and water quality

Aquaponics couples two biological processes

ANCHOR: fish waste → nitrification → plant uptake

- Fish produce ammonia-containing waste.
- Nitrifying microbes convert ammonia to nitrite and then nitrate.
- Plants remove nutrients, while water returns to the culture system.

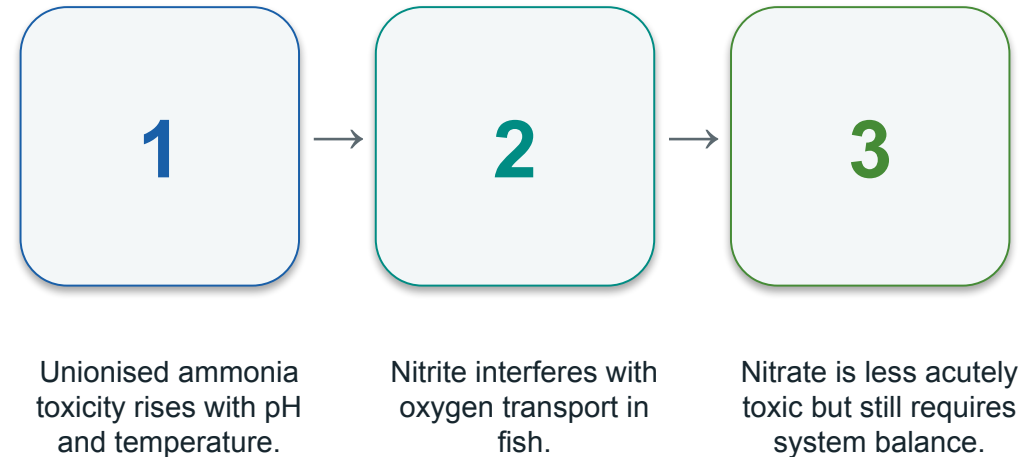


Visual: generated for this course

The nitrogen cycle has toxic intermediates

ANCHOR: $\text{NH}_3/\text{NH}_4^+ \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$

- Unionised ammonia toxicity rises with pH and temperature.
- Nitrite interferes with oxygen transport in fish.
- Nitrate is less acutely toxic but still requires system balance.



Three living groups share one water loop

ANCHOR: fish · microbes · plants

- A setpoint suitable for plants alone may harm fish or biofilter microbes.
- Feeding rate drives nutrient loading and oxygen demand.
- Management aims for an acceptable compromise, not a single-species optimum.

1 setpoint · suitable · plants · alone · harm · fish

2 Feeding · rate · drives · nutrient · loading · oxygen

3 Management · aims · acceptable · compromise · single-species · optimum

pH is a compromise variable

ANCHOR: fish health + nitrification + nutrient availability

- Nitrification tends to acidify the system over time.
- Very high pH can increase toxic unionised ammonia fraction.
- Corrections must be gradual and compatible with all organisms.

1 Nitrification · tends · acidify · system · over · time

2 Very · high · increase · toxic · unionised · ammonia

3 Corrections · must · gradual · compatible · with · organisms

Dissolved oxygen supports fish, roots and biofilter

ANCHOR: three simultaneous oxygen demands

- Warm water and high feeding increase risk.
- Aeration failure can become an emergency.
- Independent air delivery and alarms may be justified by stock value and response time.

1 Warm · water · high · feeding · increase · risk

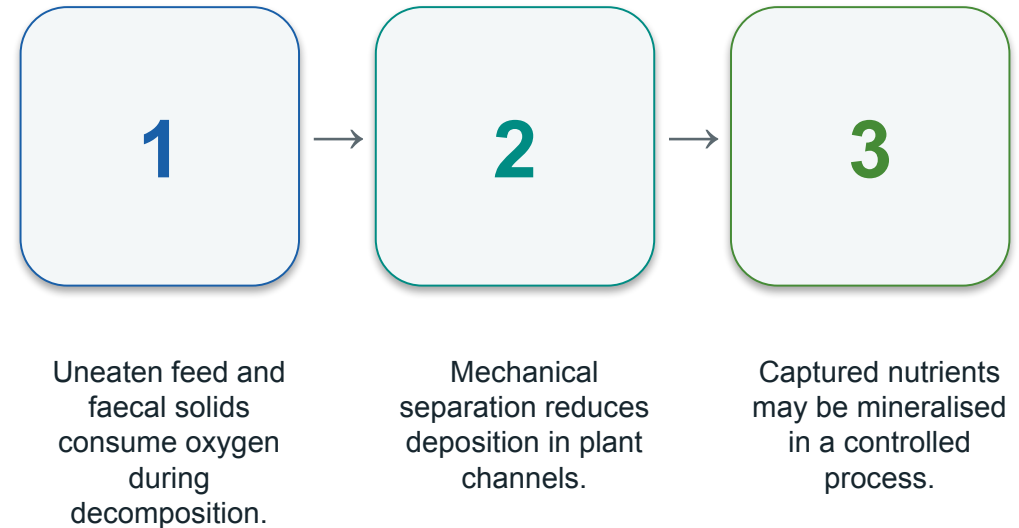
2 Aeration · failure · become · emergency

3 Independent · delivery · alarms · justified · stock · value

Solids management precedes mineralisation

ANCHOR: settle/filter solids before they foul roots

- Uneaten feed and faecal solids consume oxygen during decomposition.
- Mechanical separation reduces deposition in plant channels.
- Captured nutrients may be mineralised in a controlled process.



Biofilters need surface area and flow

ANCHOR: attached nitrifiers + oxygenated water + alkalinity

- New systems require cycling before full fish loading.
- Disinfectants and sudden chemistry changes can damage the microbial community.
- Monitor ammonia and nitrite during startup and after disturbances.

1 systems · require · cycling · before · full · fish

2 Disinfectants · sudden · chemistry · changes · damage · microbial

3 Monitor · ammonia · nitrite · during · startup · after

Feeding rate is the nutrient input

ANCHOR: feed mass/day links fish biomass to plant area

- Overfeeding raises waste, oxygen demand and water-quality risk.
- Underfeeding limits fish growth and plant nutrients.
- Use species-, feed- and system-specific design guidance.

1 Overfeeding · raises · waste · oxygen · demand · water-quality

2 Underfeeding · limits · fish · growth · plant · nutrients

3 species- · feed- · system-specific · design · guidance

Decoupled aquaponics separates optimisation

ANCHOR: controlled transfer between fish and plant loops

- A decoupled design can protect fish from crop-specific corrections.
- It adds tanks, controls, monitoring and operating complexity.
- The value depends on production goals and operator capability.

1 decoupled · design · protect · fish · from · crop-specific

2 adds · tanks · controls · monitoring · operating · complexity

3 value · depends · production · goals · operator · capability

Biosecurity spans water, animals and plants

ANCHOR: quarantine + source control + sanitation + records

- Do not transfer water, nets or plants between systems without control.
- Fish health and food-safety obligations differ from plant-only hydroponics.
- Mortality, treatment and water-quality records support investigation.

1 transfer · water · nets · plants · between · systems

2 Fish · health · food-safety · obligations · differ · from

3 Mortality · treatment · water-quality · records · support · investigation

Hydroponics and aquaponics solve different constraints

ANCHOR: formulated nutrients versus biological nutrient generation

- Hydroponics offers direct nutrient formulation and easier crop isolation.
- Aquaponics creates coupled biological value and coupled risk.
- Choose based on production objective, market, regulation and operational skill.

1 Hydroponics · offers · direct · nutrient · formulation · easier

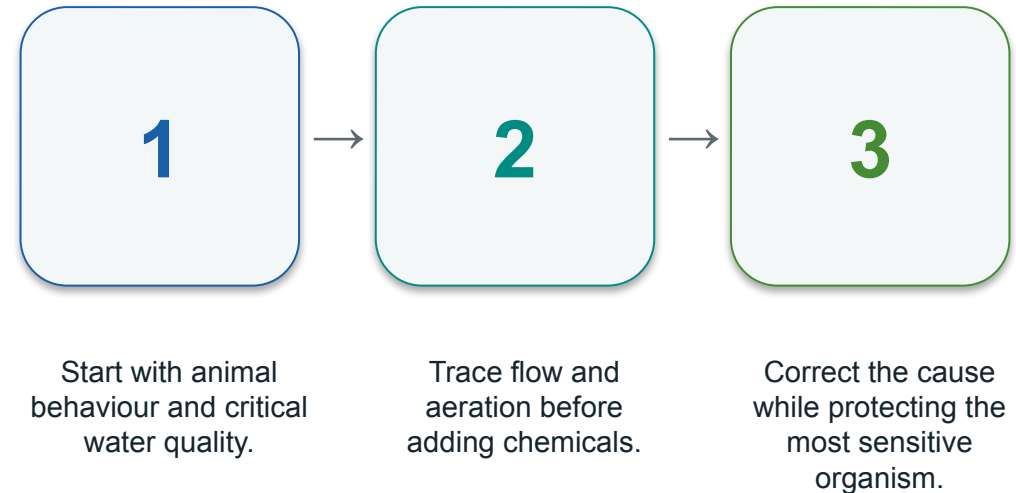
2 Aquaponics · creates · coupled · biological · value · coupled

3 Choose · based · production · objective · market · regulation

Aquaponics troubleshooting follows the loop

ANCHOR: fish → feed → oxygen → ammonia → biofilter → plant

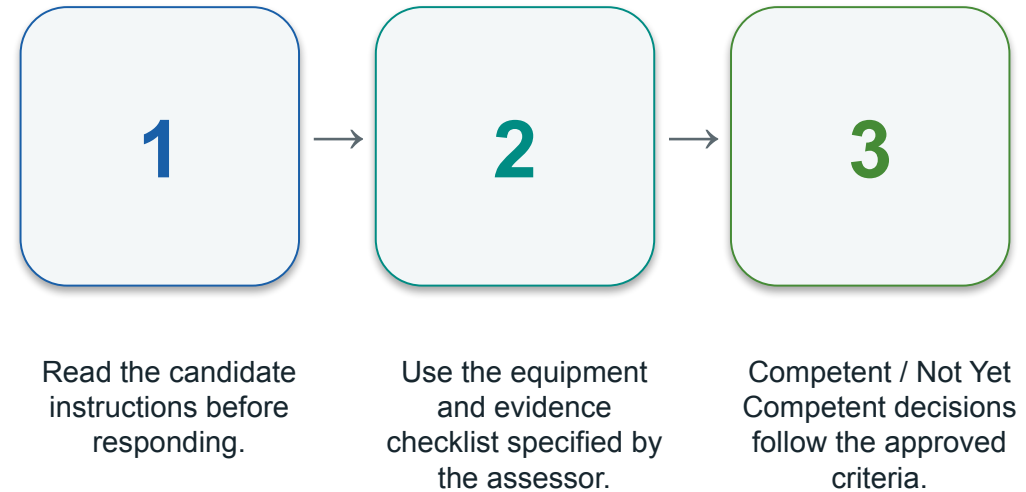
- Start with animal behaviour and critical water quality.
- Trace flow and aeration before adding chemicals.
- Correct the cause while protecting the most sensitive organism.



Assessment briefing

ANCHOR: WA-SAQ 30 min + PP 30 min

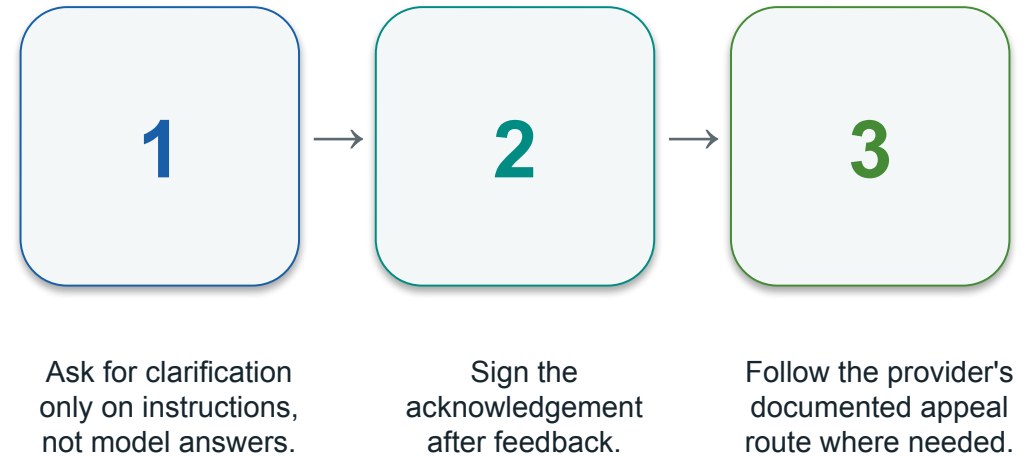
- Read the candidate instructions before responding.
- Use the equipment and evidence checklist specified by the assessor.
- Competent / Not Yet Competent decisions follow the approved criteria.



Assessment and appeal flow

ANCHOR: attempt → assessor decision → feedback → reassessment/appeal

- Ask for clarification only on instructions, not model answers.
- Sign the acknowledgement after feedback.
- Follow the provider's documented appeal route where needed.



Digital attendance and TRAQOM

ANCHOR: identity + attendance + feedback records

- Complete attendance using the provider's authorised channel.
- Do not share login or Singpass credentials.
- Submit course feedback after the assessment process.

1 Complete · attendance · using · provider's · authorised · channel

2 share · login · Singpass · credentials

3 Submit · course · feedback · after · assessment · process

Thank you

ANCHOR: control the root zone, verify the evidence

- Observe the whole water–root–canopy loop.
- Operate to SOP and stop when controls are missing.
- Use measurements and records to improve the next crop cycle.

1 Observe · whole · water–root–canopy · loop

2 Operate · stop · when · controls · missing

3 measurements · records · improve · next · crop · cycle